1941CMWCI.638....1A

Contributions from the Mount Wilson Observatory, No. 638
Reprinted from the *Astrophysical Journal*, Vol. 93, pp. 11–23, 1941
Printed in the U.S.A.

SOME RESULTS WITH THE COUDÉ SPECTROGRAPH OF THE MOUNT WILSON OBSERVATORY

WALTER S. ADAMS

ABSTRACT

The development of the coudé spectrograph in use with the 100-inch telescope is traced through its successive stages to its present form. At present three Schmidt cameras with focal lengths of 32, 73, and 114 inches, respectively, may be used in conjunction with a concave off-axis collimating mirror and a Wood aluminium on pyrex-glass plane grating. The grating has great concentration of light in the red of the first order and the ultraviolet of the second. The linear scale of the 20-inch spectrograms in the second order of the 114-inch Schmidt camera is 2.9 A/mm.

Illustrations of the application of the spectrograph consist of five investigations. The first of these is a determination of the solar parallax from 37 spectrograms of α Boötis. The range among the observations is 0.398 km/sec, and the resulting parallax is 8″.805 ± 0″.007. The radial velocity of the star is −5.621 ± 0.005 km/sec.

Seven spectrograms of Mars in the region about λ 7000 have been used in a study of the water-vapor lines. Four of the spectrograms were taken when Mars was approaching the earth, and three when it was receding. Measurements of the wave lengths of the water-vapor lines show no certain displacements from their normal terrestrial positions and indicate an upper limit for the amount of water vapor in the equatorial regions of Mars certainly not exceeding 5 per cent of that in the earth's atmosphere.

A high-dispersion spectrogram of o Ceti taken about one month after maximum of light provides evidence that the multiple structure of at least some of the emission lines of hydrogen is due to superimposed absorption lines of other elements. Comparison of the displacements of a large number of absorption lines of different excitation levels indicates a small systematic difference with level, lines of the lowest excitation level showing the smallest positive displacement.

The doubling of a considerable number of lines in the spectra of supergiant M-type stars was discovered with the coudé spectrograph. The separation and the relative intensities of the two components vary greatly in different stars, the separation being largest and the difference of intensity greatest in α Orionis, and less in α Scorpii, α^1 Herculis, and o Ceti. All the double lines arise from near the ground state of the neutral or ionized atom, and the amount of widening of other lines appears to depend upon their excitation levels. The position of the mean of the two components agrees well with that of the normal line, but the effect can hardly be ascribed to reversal and is probably due to Doppler shifts in the extensive atmospheres of these stars, as has been suggested by Spitzer.

The original discovery with the coudé spectrograph of several previously unobserved sharp interstellar lines in the ultraviolet and violet spectra of some stars of early type has been supplemented by recent observations of the bright star ζ Ophiuchi with higher dispersion. A number of additional lines has been found; and the presence of several lines of *CH*, predicted by McKellar, has been established. The existence of molecules of *CH* in interstellar space, first suggested by Swings and Rosenfeld, and given much greater probability by the work of McKellar, and of *CN* as suggested by McKellar, now seems to be beyond reasonable doubt.

The spectrograph used at the coudé focus of the 100-inch telescope has undergone many modifications since it was first placed in operation. Originally designed as an autocollimating prismatic spectrograph with an aperture of 6 inches and a focal length of 15 feet, successive developments have included the addition of a smaller instrument 9 feet in focal length with an ultraviolet glass prism, a similar spectrograph with a crystalline quartz prism aluminized on its back surface, a plane-grating autocollimating spectrograph of 9 feet focal length, and a 15-foot concave-grating spectrograph. Some of these instruments were more or less experimental in character, while others were used over considerable periods of time. Since exposure times necessarily were long, an attempt to increase the light-efficiency of these spectrographs and the extent of spectrum in focus on the photographic plate was a primary consideration. Still more important were the factors of definition and resolving-power, stability and accurate temperature control.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

The development of the Schmidt camera for spectrographic use and the remarkable success attained by Professor R. W. Wood in the ruling of diffraction gratings on aluminium surfaces with high concentration of light in the first or second orders led to a complete reconstruction of the spectrograph in 1935. A massive steel frame, resting on a three-point ball support and held at the proper angle by brackets attached to the side of the concrete pier, carries the optical parts of the spectrographs. Previously these had been attached directly to the pier and were affected by any strains or deflections in the pier itself. A structural-steel framework, triangular in shape, is attached to the I-beams of the main frame, the upper surface being horizontal. The whole structure forms approximately an inverted right triangle with the slit lying near the acute vertex and the hypothenuse parallel to the optical axis of the beam from the telescope. The horizontal base carries the various Schmidt cameras as well as a lens of 108 inches focal length. The light passing through the slit falls upon a concave collimating mirror corrected to operate off its axis. This mirror throws the parallel beam of light backward and slightly upward in the general direction of the slit to the plane grating placed just above the optical axis. By simple rotation of the grating the spectrum can be thrown, either horizontally into the Schmidt cameras and 9-foot lens camera, or downward nearly along the optical axis.

Two Schmidt cameras have been used, one with a focal length of 32 inches and the other of 73 inches. The correcting plates for these instruments and the lens of the 108-inch camera rest on a steel plate immediately in front of the grating and are readily interchangeable. The 32-inch camera and the plateholder frame for the 108-inch lens camera are mounted upon rails and can quickly be moved to one side when the 73-inch Schmidt camera is to be used. Since all the instruments are in adjustment, the change from one to the other in the course of the observations requires not more than 2 or 3 minutes. The steel frame carrying the spectrographs is completely inclosed in an insulated wooden house, and the temperature is controlled by a series of heating coils operated by an ether thermostat. Since the contacts in the thermostat are made in a hydrogen atmosphere, no oxidation occurs, and they have been used for several years without requiring attention. Stratification of air is prevented by fans placed within the wooden housing.

At the slit, provision is made for a comparison spectrum from an electric arc or vacuum tubes, a slotted diagonal prism allowing exposures to be made at any time without interrupting the exposure on the star. The optical projection system, including diffusion screen, projecting lens, and field lens over the slit, is made of quartz, so that the ultraviolet spectrum is transmitted freely. Calibration is afforded by a series of spectra from a step slit, the light from an incandescent lamp, adjustable in distance, passing through the spectrograph on either side of the comparison spectrum. A complete unit, consisting of a slit equipped with Bowen's "image-slicer" device, may be used in place of the regular slit, the change being made quickly and easily through the use of dowel pins which define the positions of both slits accurately. When the image-slicer is in use, cylindrical lenses attached to the various plateholder frames of the Schmidt cameras are brought into position in front of the photographic plate.

The most recent and an extremely valuable addition to the coudé spectrograph is the large Schmidt camera with a focal length of 114 inches. This instrument lies along the main structural frame of the spectrograph nearly in the optical axis of the telescope; and the mirror, 36 inches in diameter, is supported on a strong steel framework attached to the heavy I-beams. The mirror is the property of the Friends of Astrophysical Research and is loaned for use in this spectrograph. Its aperture permits a region of spectrum nearly 17 inches in length to receive full illumination at the plateholder, and the operating ratio of the camera is such that no correcting plate is required. As a result there is no glass or quartz in the entire optical system between the star and the photographic plate; and since all reflections are at aluminium surfaces, transmission in the ultraviolet

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941MNRAS.101..638....1A

is exceptionally high. An important practical advantage is the possibility of using ultraviolet comparison lines of higher orders in regions of the spectrum where the normal iron lines are few in number. A comparison of spectra taken with the 108-inch lens camera and the 114-inch Schmidt camera shows essentially identical definition and resolving-power, but in the case of the Schmidt camera a reduction of roughly 50 per cent in exposure time in the blue, and, of course, far higher ultraviolet transmission.

The diffraction grating used almost entirely for these various instruments is a Wood aluminium on pyrex-glass grating with a ruled area of $4\frac{3}{8} \times 5\frac{1}{2}$ inches, 14,400 lines to the inch. It has excellent concentration of light in the red-infrared of the first order and similar concentration in the violet-blue of the second order. With the cameras used, the definition is excellent in the first, second, and third orders. An illustration of its resolving-power on emission lines is the complete photographic resolution even on fast plates of the components of the *Fe* line λ 3918 with a separation of 0.099 A. The only serious drawback of this grating is the presence of strong scattered light in the far ultraviolet, and below λ 3300 a diaphragm covering about one-half its surface must be used to insure the best results.

For convenience of reference the following table gives the linear dispersion of the spectra in angstroms per millimeter for the various instruments:

32-inch Schmidt, second order	1 mm	= 10.4 A
73-inch Schmidt, second order	1	= 4.5
108-inch lens, second order	1	= 3.1
114-inch Schmidt, second order	1	= 2.9

Because of the distribution of brightness in the grating, nearly all spectrograms in the blue and violet regions have been made in the second order, and those in the red and yellow in the first order. The linear scale in the first order is, of course, closely one-half that of the second.

Investigations carried on with the coudé spectrographs have naturally been limited to the brighter stars. The faintest star so far observed with the highest dispersion (114-inch Schmidt) is *o* Ceti when it had a photographic magnitude of about 6.0. The exposure time was $5\frac{1}{2}$ hours. With the 32-inch Schmidt camera stars of magnitude 7.0 have been observed frequently, and fainter stars to a less extent. In the yellow and red region of the spectrum, the number of reflections from aluminium surfaces (six including the telescope and spectrograph) results in a serious loss of light as compared with that from silver surfaces, and observations are necessarily limited to very bright stars or those of late spectral type.

The use of high resolving-power and large linear scale is most valuable for accurate determinations of radial velocity, the detection of differential displacements among lines of various elements, physical studies of line-structure and contours, and the discovery of faint lines and bands. Illustrations of some of these applications are contained in the remaining pages of this communication.

α BOÖTIS AND THE SOLAR PARALLAX

The lines in the spectrum of α Boötis remain sharp and narrow under the highest dispersion applied to the study of this star. Accordingly, it has been used frequently by astronomers as a star of standard velocity to serve as a control upon the adjustments and behavior of their spectrographs. Its brightness and convenient position have made it equally useful in connection with the coudé spectrograph; and during the past years many spectrograms have been obtained—37 of these since the grating came into regular use in the spectrograph. Since these spectrograms were taken under reasonably similar conditions, it seemed of interest to attempt a solution for the solar parallax with this comparatively limited material with a view to learning what degree of accuracy might be expected of the method in the future.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

The spectrograms were all taken in the blue and violet region of the spectrum in the second order of the grating. The first 28 were with the 108-inch lens camera, and the last 9 with the 114-inch Schmidt. Since there seems to be no appreciable systematic difference between the results with the two cameras, all the material has been used. The plates are given equal weight and are used individually except that the two spectrograms, taken in immediate succession on March 2, 1939, are combined into a single value with unit weight. Cramer Hi-Speed Special plates were used for nearly all the spectrograms with exposure times of 30–40 minutes under average conditions of seeing.

A list of 25 suitable lines in the region between λ 4337 and λ 4466 was selected for measurement; and in so far as possible, the same lines were measured upon all the negatives. The wave lengths of the stellar lines were taken from the *Revised Rowland Table*, and those of the comparison lines from the measures of St. John and Babcock of the iron spectrum in the laboratory. Although the absolute radial velocity derived in this way will differ from that in which the same wave lengths are used for both stellar and comparison lines to the extent to which solar wave lengths differ systematically from terrestrial wave lengths, solar wave lengths have the advantage of being strictly homogeneous. In a differential comparison, like the present one, the system of wave lengths used is, of course, of no importance.

Table 1 contains the values for the individual spectrograms. The successive columns give the number of the plate; the date in Greenwich Civil Time; v_0, the correction due to the earth's orbital motion; v_d, the correction due to the earth's rotation; and V, the final radial velocity of the star. If v_m is the directly measured radial velocity, $V = v_m + v_0 + v_d$.

The method for determining the solar parallax from measurements of the radial velocity of one or more stars at various epochs has been worked out in detail and applied by several observers[1] and hence does not require extended comment. For simplicity, a standard value of the parallax is assumed, and a correction $d\pi$ is derived by least squares from the equations of condition provided by the observations. In the present case there are 37 equations of condition of very simple form, since equal weights are assumed. The solution on the basis of an assumed parallax of 8″.80 gives $d\pi = +0''.0054$. The final results, with their probable errors, are

$$\pi = 8''.805 \pm 0''.007\,, \qquad V = -5.621 \pm 0.005 \text{ km/sec}\,.$$

Although these results are fairly satisfactory in view of the relatively small amount of observational material, the probable error is larger than it should be, considering the small probable error of the determination of V from a single spectrogram. The extreme range in values of the 25 lines measured rarely exceeds 1.5 km/sec, and the probable error of the mean averages ±0.05 km/sec. The values of V given in Table 1 show larger differences among the separate plates than would be expected under these circumstances, and it is clear that slight systematic sources of error must be present. These are probably to be sought for in the character of the arc which produces the comparison spectrum and in the general system of temperature control throughout the spectrograph. These are, at present, under investigation. If the sources of systematic error can be removed or even considerably reduced, it is evident that a highly accurate value of the solar parallax can be derived at the cost of a small fraction of the labor required in observation and computation by the more usual methods.

The radial velocity of α Boötis derived from numerous sources and given in Moore's *Catalogue of Radial Velocities*[2] is −5.2 km/sec. Since most of the results were obtained with the use of the same wave lengths for comparison and stellar lines, it is necessary to make allowance for an average difference of +0.006 A (solar − comparison) in comparing the coudé results with previous determinations. This would give a value of −5.2 km/sec,

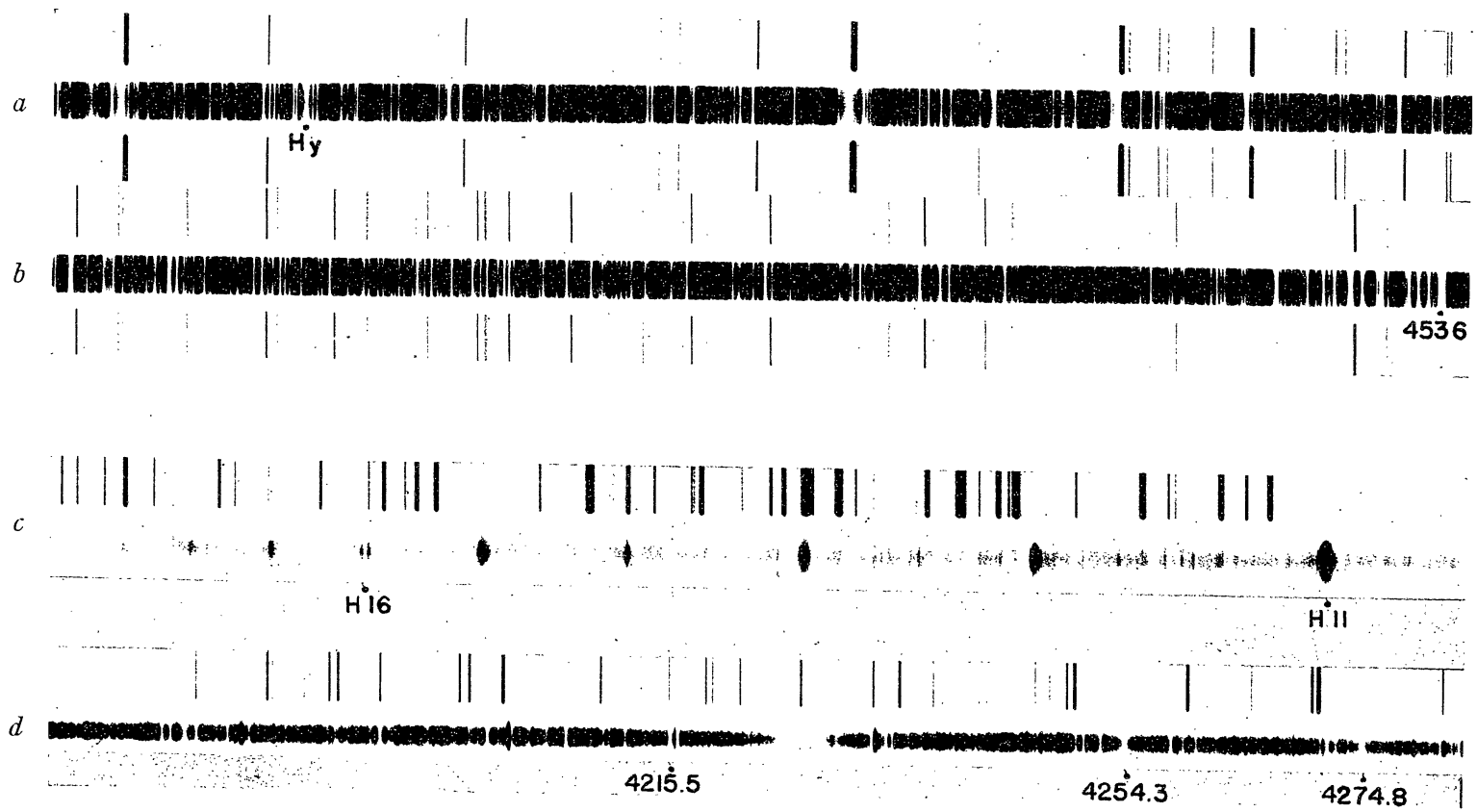
[1] Among others, H. Spencer Jones, *Annals Cape Obs.*, **10**, Part 8, 1928.

[2] *Pub. Lick. Obs.*, **18**, 1932.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

PLATE V



STELLAR SPECTRA WITH HIGH DISPERSION

a, *b*) α Boötis; *c*, *d*) *o* Ceti. Note structure of *H*16 and doubling of λ 4254 and λ 4274

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1991CWA1.638....1A

as against −5.2 for the mean of the collected results, or −5.4 for the Lick Observatory observations alone.

An interesting illustration of differences in wave length between solar and stellar lines is afforded by the *V* line λ 4428 in the spectrum of α Boötis. This is listed as

TABLE 1

RADIAL VELOCITIES OF α BOÖTIS

Plate No.	Date	v_0	v_d	V
		km/sec	km/sec	km/sec
1356	1937 Apr. 25.23	− 4.568	+0.125	−5.606
1360	26.25	− 5.004	+ .166	−5.532
1370	May 27.45	−16.806	− .353	−5.431
1371	28.18	−17.037	+ .122	−5.590
1375	29.17	−17.344	+ .137	−5.434
1399	June 27.16	−23.861	− .014	−5.554
1402	28.18	−23.994	− .063	−5.734
1405	29.15	−24.114	− .013	−5.782
1425	July 27.18	−24.736	− .238	−5.668
1596	Dec. 22.55	+23.944	+ .257	−5.667
1618	1938 Jan. 20.51	+25.830	+ .162	−5.570
1628	Mar. 15.48	+12.869	− .098	−5.663
1633	16.46	+12.486	− .057	−5.789
1635	16.51	+12.466	− .170	−5.828
1670	May 14.28	−12.153	− .008	−5.832
1894	1939 Jan. 10.53	+25.929	+ .197	−5.488
1897	11.50	+25.955	+ .237	−5.498
1901	30.51	+24.958	− .019	−5.519
1913	Mar. 1.50	+17.977	− .057	−5.436
1916–17	2.44	+17.660	+ .011	−5.591
1983	Apr. 8.36	+ 2.985	+ .038	−5.494
1984	8.41	+ 2.961	− .087	−5.776
1990	9.34	+ 2.557	+ .074	−5.829
1993	27.25	− 5.208	+ .156	−5.700
1996	28.28	− 5.644	+ .093	−5.818
2030	June 1.17	−18.099	+ .143	−5.429
2033	2.15	−18.387	+ .168	−5.763
2036	3.15	−18.674	+ .165	−5.510
2052	29.16	−24.054	− .013	−5.417
2056	30.15	−24.173	.000	−5.726
2060	July 1.15	−24.287	− .013	−5.777
2080	Aug. 24.18	−20.078	− .330	−5.490
2084	25.13	−19.830	− .271	−5.664
2224	Dec. 31.53	+25.182	+ .246	−5.464
2230	1940 Jan. 19.52	+25.898	+ .193	−5.796
2276	Mar. 20.37	+10.732	+ .130	−5.640
2281	21.49	+10.275	−0.154	−5.464

4428.551 *V—d*? in the *Rowland Table*. It is of excellent quality in α Boötis but gives a systematic difference of −2.0 km/sec from other stellar lines. Mr. Babcock made a special study of this line when it was called to his attention and found no less than three lines apparently involved in the solar wave length: 4428.51 *Cr*, 4428.52 *V*, and 4428.55

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

Fe predicted. On a photograph taken in the third order of the 75-foot spectrograph the last two lines were resolved, and their wave lengths measured as 4428.524 and 4428.578. It is clear that in α Boötis the *V* line is the effective one and that a wave length of 4428.524 gives good agreement with the results from other lines.

WATER-VAPOR LINES IN THE SPECTRUM OF MARS

Four spectrograms of Mars in the region covering the B band of oxygen and the water-vapor lines as far as λ 7300 were taken during February–April, 1937, and three during October–December, 1939. The first four photographs were taken on IIIN

TABLE 2

MEASUREMENTS OF SPECTROGRAMS OF MARS

Plate No.	Calc. *V*	Obs. *V*	Displacements (Rowland λ)		Displacements (Babcock λ)	
			A_{wv}	*A*	A_{wv}	*A*
	km/sec	km/sec	A	A	A	A
1329.....	−18.9	−19.4	+0.051(12)	+0.005 (9)	+0.022 (2)	+0.002(10)
1337.....	−18.7	−18.9	+ .005 (6)	+ .019(13)	+ .019 (6)	+ .020(12)
1338.....	−16.7	−17.7	+ .020(11)	− .012(12)	+ .023(11)	+ .003 (9)
1361.....	−11.5	−11.4	+ .011 (3)	+ .006 (9)	− .002 (3)	− .002(10)
2143.....	+12.9	+12.5	+ .012(13)		+ .027(14)	
2170.....	+15.1	+14.8	+ .008 (9)	+ .019 (9)	+ .017 (9)	+ .016(10)
2177.....	+15.5	+15.5	+0.026 (3)	+0.017(23)	+0.027 (3)	+0.018(27)

TABLE 3

MEAN DISPLACEMENTS

Plate Nos.	Rowland λ		Babcock λ	
	A_{wv}	*A*	A_{wv}	*A*
1329–61.....	+0.018 A	+0.005 A	+0.018 A	+0.007 A
2143–77.....	+0.012	+0.018	+0.024	+0.017

emulsions, one of the others on IS and the other two on IN. They were made in the first order of the 9-foot lens spectrograph and are of only fair quality, the aluminized surfaces of the mirrors in both telescope and spectrograph causing a serious loss of light in this region of the spectrum, as compared with silver surfaces. The coarseness of the grain of the IN emulsions is also a drawback to accurate measurement. The series of spectrograms, however, has the important advantage of having been taken in part when Mars was approaching the earth and in part when it was receding, so that Martian components of the water-vapor lines should lie on opposite sides of the terrestrial lines, and the problem of their detection by measurement becomes a differential one.

Since the comparison lines of iron in this region of the spectrum are, for the most part, faint and few in number, the sharp lines of oxygen in the B band have been used as standards for the wave lengths of the water-vapor lines. It has been shown previously[3]

[3] *Mt. W. Contr.*, No. 488; *Ap. J.*, **79**, 308, 1934.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1940PASP..52..341A

that the amount of free oxygen in the atmosphere of Mars which could produce components of these lines must be extremely small, and the use of the terrestrial wave lengths of these lines will show whether or not the water-vapor lines behave similarly. A check on the procedure is also afforded by the displacements of the lines of solar origin, which should, of course, show displacements corresponding to the theoretical radial velocity of Mars with reference to the earth, i.e., the time-rate of change of the light-path sun to Mars to earth.

In Table 2 are given in successive columns the number of the spectrogram, the calculated radial velocity of Mars, the observed radial velocity based upon the wave lengths of the oxygen lines as standards, and the displacements in angstroms of two groups of lines designated in the *Revised Rowland Table* as A_{wv} and A, respectively. It seems very probable that both sets of lines are due to water vapor. Two sets of wave lengths are used for the solar, oxygen, and water-vapor lines—the first, those of the *Rowland Table*, and the second, more recent determinations by Babcock. The number of lines is given in parenthesis.

If the results for the first four spectrograms when Mars was approaching are compared with the last three when it was receding, the differences between calculated and observed velocity, −0.45 km/sec and −0.23 km/sec, differ so little from each other as to afford no evidence of change in the position of the oxygen lines upon which the determinations are based. In fact, the sign of the difference is in the wrong direction for the presence of a Martian component to the oxygen lines. Similar means for the groups of A_{wv} and A lines, weighted according to the number of lines, are given in Table 3.

The persistence of the positive sign in these differences is very marked. It may be due to a systematic difference in wave length between the oxygen and the water-vapor lines, which is not at all probable; or to the reduction of the wave lengths of the water-vapor lines on the spectrograms from the oxygen lines, which involves some extrapolation, although the attempt was made to allow for this factor. The important question, however, is that of the differences between the two groups of values in Table 3. When the results for the two sets of wave lengths are combined, the difference, Mars approaching *minus* Mars receding, is 0.000 for the A_{wv} lines, −0.012 for the A lines, or −0.006 A for the mean of the two. This quantity, although of the right sign to indicate the presence of Martian lines, is of little or no significance when the accidental error of measurement of these rather diffuse lines is considered. The general conclusion to be drawn from this investigation with rather limited material is that, if water-vapor lines are present in the spectrum of the equatorial areas of Mars, they cannot be more than 5 per cent as strong as in the earth's atmosphere and are probably very much less.

I am indebted to Dr. Dunham for some of the spectrograms used in this investigation.

THE SPECTRUM OF *o* CETI

A large-scale spectrogram of *o* Ceti with the 114-inch Schmidt camera was obtained on September 29, 1939, in the second order of the grating. The maximum of light had occurred in August, and the star's visual magnitude at the time of observation was about 4.2. The photograph covers the region λλ 3700–4900 and is of good quality to show the absorption spectrum and bands. The bright hydrogen lines $H\zeta$, $H\delta$, and $H\gamma$ are necessarily overexposed, but other fainter emission lines are well shown. The exposure time was $5\frac{1}{2}$ hours on a Cramer Hi-Speed Special plate.

The resolving-power of this spectrogram is such that the study of the molecular bands forms an extensive investigation in itself, which is being undertaken by Miss Dorothy N. Davis in connection with similar work on other M-type stars. Certain features of the atomic spectrum have also proved of sufficient interest to deserve comment.

A simple inspection of the spectrum shows that the cause producing great widening, and in numerous cases doubling, of lines in the spectra of *α* Orionis, *α* Scorpii, *α* Herculis,

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

and other M-type supergiants is present in *o* Ceti, although in less degree. The lines of the *Mn* I triplet near λ 4030 and λ 4275 of *Cr* I are resolved, and their separations have been measured, while the other two lines of the *Cr* I triplet, the lines of *Sr* II λ 4077 and λ 4215, and the *Ba* II line λ 4554 show great widening but are not resolved. *Ca* I 4227 is not measurable. All these lines arise from the ground level of the neutral or ionized atom. Measures of the separations in *o* Ceti are given in a later section of this communication where M-type stars are discussed. Many of the emission lines in this spectrum appear as bright wings on the violet side of the corresponding absorption lines. In a few cases there seems to be evidence of similar bright wings on the red side, but this is by no means conclusive. They may be portions of the continuous spectrum or close lines of other origin, but they usually appear only in connection with strong absorption lines. Illustrations are λλ 3893, 3907, 3956, and 3992. An interesting fact about the emission lines appearing on the violet side of absorption lines is that apparently they do not affect the wave lengths of the absorption lines. For example, the well-known *Fe* I lines λ 4202 and λ 4376 have strong emission components, but the absorption lines have the same displacements as other absorption lines unaccompanied by emission. Furthermore, the displacements of such emission lines correspond closely to those of emission lines unaccompanied by absorption, such as the hydrogen and silicon lines. These results argue strongly against reversal and for an independent spectrum of bright lines superimposed upon the absorption spectrum.

The behavior of the emission lines of hydrogen in the spectrum of *o* Ceti has always presented a remarkable problem. Many of the lines of the Balmer series are wide but single, while others, notably *Hγ*, *Hδ*, *Hζ*, *H*10, and *H*16, show two, three, or even more components. It is exceedingly difficult to ascribe such a diversity of behavior of the series lines of a single element to intrinsic causes alone, and Merrill has suggested that in the lines showing components the effect is merely an apparent one due to the superposition of absorption lines upon the broad emission lines, thus dividing them into what appear to be individual lines. That this occurs in *H*16 is quite certain. *H*16 is a double bright line with strong central absorption due to the *V* line λ 3703.59 of low excitation.[4] Its position agrees perfectly with that of other *V* lines of similar excitation in the general absorption spectrum. *H*10 appears on the high-dispersion spectrogram as a triplet with relatively faint absorption lines or spaces dividing the components. Measures of the positions of these absorption features show that they can be identified with the lines λ 3797.52 *Fe* 5 and λ 3797.72 *Cr* I of the solar spectrum. Their displacements agree closely with those of other absorption lines in their vicinity; and, although their identification is not so certain as that in the case of *H*16, it seems fairly probable.

The emission lines *Hζ*, *Hδ*, and *Hγ* are greatly overexposed upon the present spectrogram, but some details have been measured upon other spectrograms taken by Joy with less exposure. A faint absorption feature dividing *Hζ* into nearly equal parts agrees closely in wave length with λ 3888.83 *Fe* 2 in the solar spectrum, and a strong absorption line forming the violet edge of the strong emission is λ 3888.52 *Fe* 5. The *Hδ* line is characterized by an apparent emission component on the red side. Whether this is a portion of *Hδ* set off by absorption or is due to *V* 4102.17 is uncertain, since but few *V* lines appear in emission in the spectrum. The apparent absorption setting off this component may be λ 4101.68 *Fe* 3 in the solar spectrum, which is nearly coincident with the hazy hydrogen line itself. At *Hγ* a fairly strong emission line lies just to the red of the principal line. It almost certainly forms a part of *Hγ*, although it may be influenced by a possible violet component on the strong *V* line λ 4341. It is separated from the main *Hγ* line by an absorption feature which corresponds to no line of the *Rowland Table*. A spectrum of α Herculis, however, shows a maximum of absorption on the red edge of the wide hazy *Hγ*, which would coincide reasonably well with the absorption in *o* Ceti.

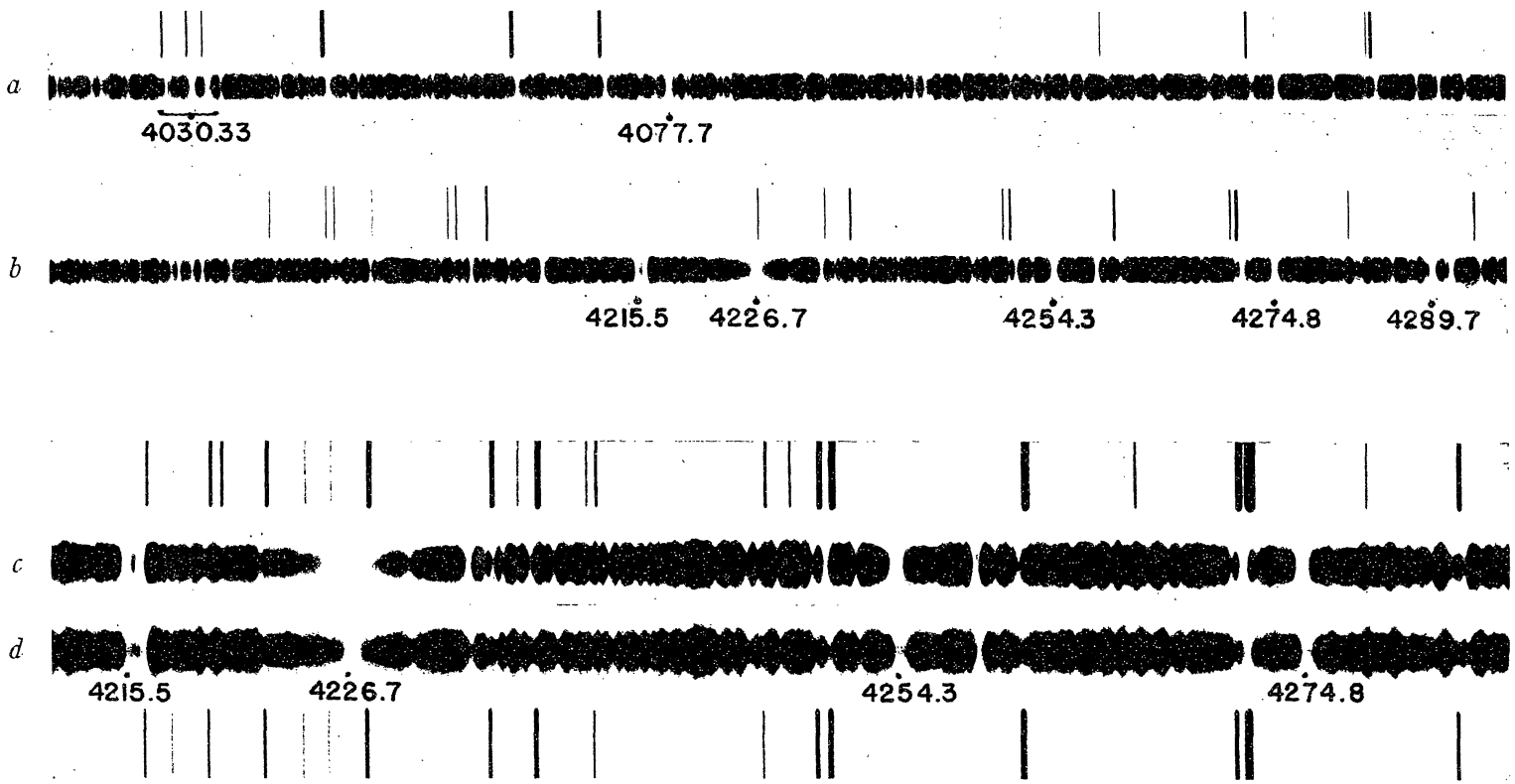
[4] C. D. Shane had previously suggested (*Lick Obs. Bull.*, **10**, 131, 1922) that this *V* line might be the cause of the abnormal faintness of *H*16.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

PLATE VI



STELLAR SPECTRA WITH HIGH DISPERSION

a, *b*) α Orionis. Note doubling of λλ 4030–33, 4077, 4215, 4226, 4254, 4274, and 4289

c) α Herculis; *d*) α Scorpii

1941CMWCI.638....1A

The general conclusion from these results is that, while superposed absorption lines may not account fully for the multiple character of several emission hydrogen lines in *o* Ceti, they must affect them greatly. In the case of *H*16 the evidence is conclusive and in that of *H*10 very probable. These results lend support to Merrill's view that the absorption spectrum in variable stars of this type is produced in a portion of the stellar atmosphere overlying that which gives rise to the emission lines.

Many absorption lines in the spectrum of *o* Ceti have been measured with a view to the study of possible differential displacements. If the lines are divided into groups according to their lower excitation potentials, a slight but systematic effect seems to be present. The results are given in Table 4. *V* is the mean velocity given by each group with its probable error. Although the differences are small, the quality of the lines and the size of the probable errors afford considerable evidence for the reality of the effect.

TABLE 4

DISPLACEMENTS OF LINES IN THE SPECTRUM OF *o* CETI

Low E.P.	Mean Low E.P.	No of Lines	*V*
			km/sec
0.00–0.19	0.04	51	+60.96±0.20
0.20–1.00	0.64	37	+62.59±0.22
1.01–1.99	1.45	34	+63.33±0.29
>2.00	2.43	32	+64.10±0.35

DOUBLE LINES IN M-TYPE STARS

The double character of numerous lines in the spectrum of α Orionis was first recognized on a spectrogram taken with the grating in December, 1936. Previously a marked asymmetry of these lines had been observed with the prismatic spectrograph, and an extensive investigation of the effect with the aid of the microphotometer has been completed and published by Dr. Lyman Spitzer.[5] Other supergiant stars of the M type showing doubling to some extent, but much less clearly than α Orionis, are α Scorpii and α^1 Herculis. Apparently ordinary giant M stars like β Andromedae, μ Geminorum, δ and ν Virginis, and σ Librae do not show the effect, all lines being sharp. The spectrogram of *o* Ceti taken about one month after maximum light, however, shows distinct doubling of the same lines as in α Orionis.

A simple examination of the spectrum of α Orionis shows that all lines are much widened when compared with the spectrum of such a star as α Boötis, for example. Some lines are much more widened than others, however, in many cases showing traces of doubling, and finally a group of lines is found which are fully resolved into two components. In α Orionis these components are of very unequal intensity, the violet member, which is sharp and strong, having several times the intensity of the diffuse red component. An interesting fact about these double lines is that all arise from very nearly the ground state of the neutral or ionized atom. The most widened lines which are not fully resolved arise from the next lowest level of excitation potential, while lines which arise from higher levels, although widened, show no evidence of doubling. Table 5 gives a list of the resolved doublets and of lines showing evidence of doubling, though not fully resolved, on a recent spectrogram in the region λλ 3750–4600.

The list of doublets not fully resolved is certainly incomplete, but enough are given to indicate the higher average excitation potential of the lower state from which they arise. The H and K lines of *Ca* II are not resolved merely because of the great intensity

[5] *Mt. W. Contr.*, No. 619; *Ap. J.*, **90**, 494, 1939.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

and diffuseness of the components. Almost without exception the components of the resolved doublets have very unequal intensities with the violet component strong and sharp, while the components of the unresolved doublets are much more equal in intensity and more diffuse in appearance.

The separation of the components of the double lines has been measured on eight spectrograms of α Orionis, ranging in date from December, 1936, to January, 1940. The lines include most of the double lines listed in Table 5, especially the *Mn* I triplet near λ 4030, the *Sr* II pair λ 4077 and λ 4215, and the *Cr* I triplet λλ 4254, 4275, and 4289. Of the double lines in the spectrum, these are of the best quality for measurement.

TABLE 5

DOUBLE LINES IN THE SPECTRUM OF α ORIONIS

RESOLVED DOUBLETS

λ	Elem.	Low E.P.	λ	Elem.	Low E.P.
3759.30	*Ti* II	0.60	3981.78	*Ti* I	0.00
3761.32	*Ti* II	.57	3989.77	*Ti* I	.02
3824.45	*Fe* I	.00	3998.64	*Ti* I	.05
3856.38	*Fe* I	.05	4030.77	*Mn* I	.00
3859.92	*Fe* I	.00	4033.08	*Mn* I	.00
3886.30	*Fe* I	.05	4034.49	*Mn* I	.00
3895.67	*Fe* I	.11	4077.73	*Sr* II	.00
3899.72	*Fe* I	.09	4215.52	*Sr* II	.00
3920.27	*Fe* I	.12	4226.74	*Ca* I	.00
3922.92	*Fe* I	.05	4246.84	*Sc* II	.31
3927.94	*Fe* I	.11	4254.35	*Cr* I	.00
3930.31	*Fe* I	.09	4274.81	*Cr* I	.00
3944.02	*Al* I	.00	4289.73	*Cr* I	.00
3948.68	*Ti* I	.00	4554.04	*Ba* II	0.00
3961.54	*Al* I	0.00			

DOUBLETS NOT FULLY RESOLVED

λ	Elem.	Low E.P.	λ	Elem.	Low E.P.
3758.25	*Fe* I	0.95	3878.58	*Fe* I	0.95
3763.80	*Fe* I	0.99	3933.68	*Ca* II	0.00
3767.21	*Fe* I	1.01	3968.49	*Ca* II	0.00
3820.44	*Fe* I	0.86	4045.83	*Fe* I	1.48
3825.89	*Fe* I	0.91	4063.61	*Fe* I	1.55
3834.24	*Fe* I	0.95	4375.95	*Fe* I	0.00

Table 6 gives, in kilometers per second, the radial velocity from each component, the mean velocity, and the radial velocity of the star as determined from a list of selected single lines. The last two columns give the separations in kilometers per second, and in angstroms, respectively.

It is well known that the radial velocity of α Orionis is variable, and these results show that the mean of the velocities given by the two components of the doublets agrees fairly closely with the stellar velocity. The average difference of +1.4 km/sec is within the limits of error of measurement. The centers of these double lines, therefore, have very nearly their normal wave lengths. Whether the separation of the components varies on different spectrograms is uncertain; the range of nearly 7 km appears large, but the measurement of the diffuse second component is somewhat difficult.

The separations of the doublets agree with one another rather well except in the case

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

of *Ca* λ 4227. This line shows a separation larger by about 0.12 A, on the average, than the other doublets. A few measures on the incompletely resolved components of H and K also give larger separations.

In the spectra of α Herculis, α Scorpii, and o Ceti a few of the doublets measured in α Orionis appear as double lines but only partially resolved. The most prominent of these are the *Mn* I, *Sr* II, and *Cr* I lines already mentioned. The relative intensities of the components, however, are much more nearly equal than in α Orionis, and this fact makes their resolution more difficult. The average separation of the components in these stars is: α Herculis, 0.27 A; α Scorpii, 0.28 A; o Ceti, 0.19 A. These values are considerably smaller than in α Orionis.

The explanation of the doubling and great widening of numerous lines in the spectra of these M-type stars on the basis of motions in their extended atmospheres, a hypothesis discussed in detail by Spitzer, seems to be most nearly in accord with the facts of observation. No emission other than bright wings on H and K is present in the spectra of any

TABLE 6

VELOCITIES FROM DOUBLE LINES IN THE SPECTRUM OF α ORIONIS

Plate No.	Date	Comp. 1	Comp. 2	Mean	Stellar Velocity	Separation	
		km/sec	km/sec	km/sec	km/sec	km/sec	A
1289.....	1936, Dec. 23	+ 8.6	+35.5	+22.0	+22.2	26.9	0.369
1331.....	1937, Feb. 23	11.0	36.0	23.5	22.0	25.0	.342
1530.....	Oct. 21	11.5	33.4	22.4	19.9	21.9	.300
1594.....	Dec. 21	9.7	36.7	23.2	21.8	27.0	.370
1822.....	1938, Nov. 8	8.7	29.5	19.1	16.0	20.8	.285
1827.....	9	9.4	30.8	20.1	17.7	21.4	.293
1911.....	1939, Feb. 28	8.3	36.0	22.2	23.3	27.7	.379
2227.....	1940, Jan. 18	+14.3	+35.9	+25.1	+23.3	21.6	0.296

of these stars except o Ceti. The great difference in the relative intensities of the components of the double lines in α Orionis is also opposed to the view that the doubling can be an effect of reversal. The hypothesis that the difference in displacement between emission and absorption lines in the spectra of Me variables is due to Doppler motion is rather generally accepted by astronomers, and the doubling and displacements of absorption lines in the spectra of the stars here considered may well be quite analogous in character.

INTERSTELLAR LINES

The discovery of several previously unobserved narrow interstellar lines was announced by Dunham and Adams in December, 1936,[6] and the identification of four lines in the far ultraviolet with ground-state lines of *Ti* II was established. In a later communication Dunham presented evidence for the presence of interstellar *Ca* I 4227 and *K* I 7699.[7] The principal stars observed were χ^2 Orionis, 55 Cygni, and o Persei. Several prominent lines remained unidentified, notably three at λλ 3957.72, 4232.58, and 4300.33. Marked variations in the intensities of these lines relative to each other and especially to interstellar H and K were noted in different stars and have been commented upon by Dunham.

Observations of these stars were made almost wholly with the 32-inch Schmidt camera and fine-grained plates. Since the interstellar lines are extremely narrow, it

[6] *Pub. A.A.S.*, **9**, 5, 1937.

[7] *Pub. A.S.P.*, **49**, 26, 1937.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

seemed desirable to use the greater photographic resolving-power and the larger linear scale of the 114-inch Schmidt camera, and a few bright stars of type approximating B0 were selected for observation. One of these, ζ Ophiuchi, of magnitude 2.7 and type B0n, has proved to be exceptionally favorable for a study of the unidentified lines. Although interstellar H and K and the D lines are of relatively moderate strength in the spectrum, the unidentified lines λλ 3957, 4232, and 4300 are very well marked. On a spectrogram taken on a coarse-grained plate these three lines, with H and K, are the principal features. In addition, a faint narrow line has been measured at λ 3874.6. On the other hand, λ 3934 and *Ca* I 4227 do not appear. This calcium line seems to be definitely correlated in intensity with H and K, and unless these are exceptionally strong, λ 4227 is not seen.

In 1937 Swings and Rosenfeld[8] offered the suggestion that the interstellar line λ 4300.3 might be due to *CH*, and recently McKellar has applied to molecular spectra the considerations derived from the study of the interstellar atomic lines of *Ti* II in the far

TABLE 7

INTERSTELLAR LINES OF *CH*

McKellar		ζ Ophiuchi		
λ	Intensity	λ	Intensity	Δλ
3878.8	1	3878.81	1	0.0
3886.39	3	3886.44	3	+ .05
3890.23	2	3890.25	2	+ .02
4300.29	>6	4300.34	8	+0.05

ultraviolet and has succeeded in identifying with considerable probability three of the lines with lines arising from common diatomic molecules.[9] These are: λ 4300.33 with *CH*, as already suggested by Swings and Rosenfeld; λ 3934.3 with *NaH;* and λ 3874.62 with *CN*. McKellar has made these provisional identifications on the basis of agreement of wave length through consideration of the few lines which would arise from the most probable transitions for molecules in the lowest rotational level of the zero vibrational level of the ground electronic state. As a result of this investigation McKellar could predict the presence of certain fainter interstellar lines in addition to those already identified, in case the identification were correct.

An examination of the available spectrograms did not show any of these additional lines with certainty; but since the visibility of faint, narrow lines depends greatly upon the contrast and grain of the photographic plate, the attempt was at once made to obtain a suitable spectrogram with the 114-inch Schmidt camera. A Cramer Contrast plate was used, and the star ζ Ophiuchi was selected because of its brightness and the intensity of several of its interstellar lines, including λ 4300. An exposure of nearly five hours gave a fully satisfactory plate in the region λλ 3700–4700.

Since the line at λ 3934.3 ascribed to *NaH* by McKellar does not appear to be present in the spectrum of ζ Ophiuchi, it is hardly to be expected that any of the other predicted lines of this molecule would appear. The line λ 3874 ascribed to *CN* is well marked, but the predicted line λ 3580 is outside the range of the spectrogram. The chief interest lies in the predicted lines of *CH*, the strongest line of which is at λ 4300. The three additional lines given by McKellar lie at λλ 3879, 3886, and 3890, with estimated intensities of 1, 3, and 2, respectively. Their combined intensity should be somewhat less than that of

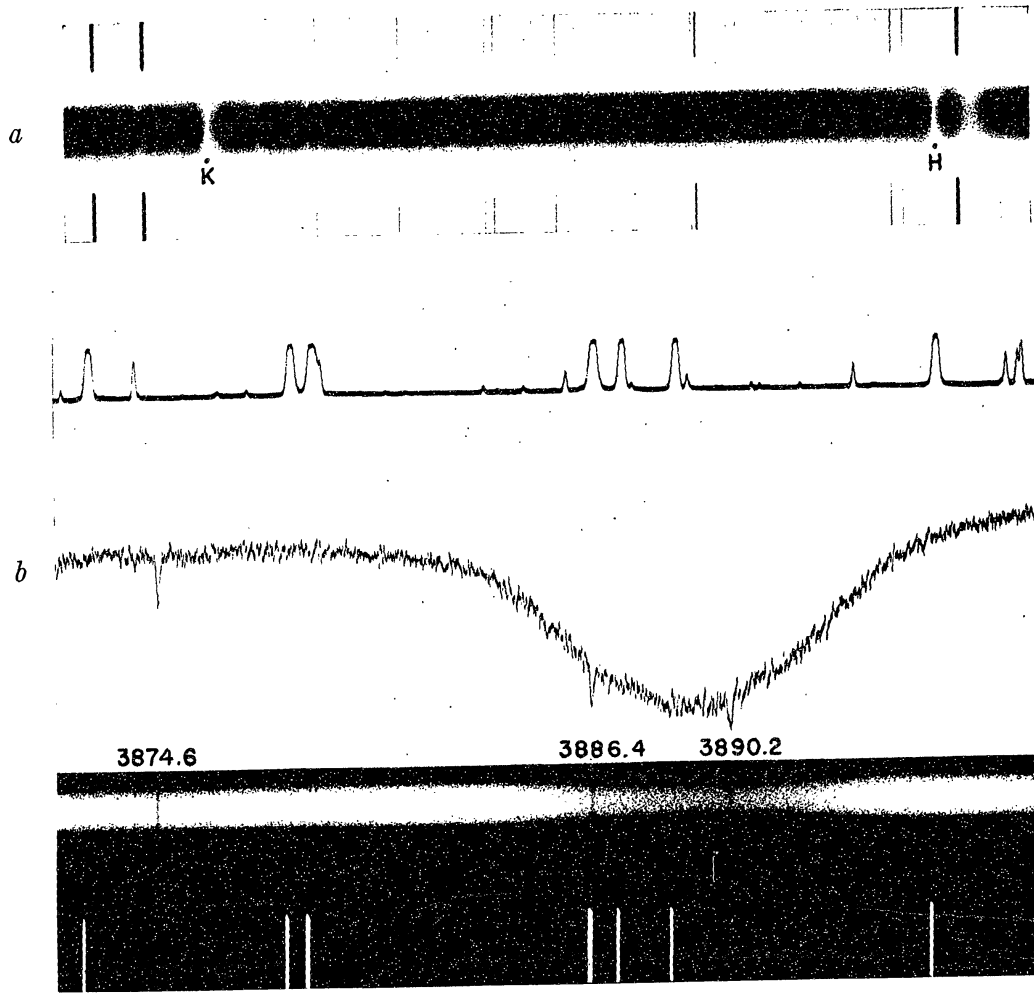
[8] *Ap. J.*, **86**, 483, 1937.

[9] *Pub. A.S.P.*, **52**, 187, 1940.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

PLATE VII



INTERSTELLAR LINES

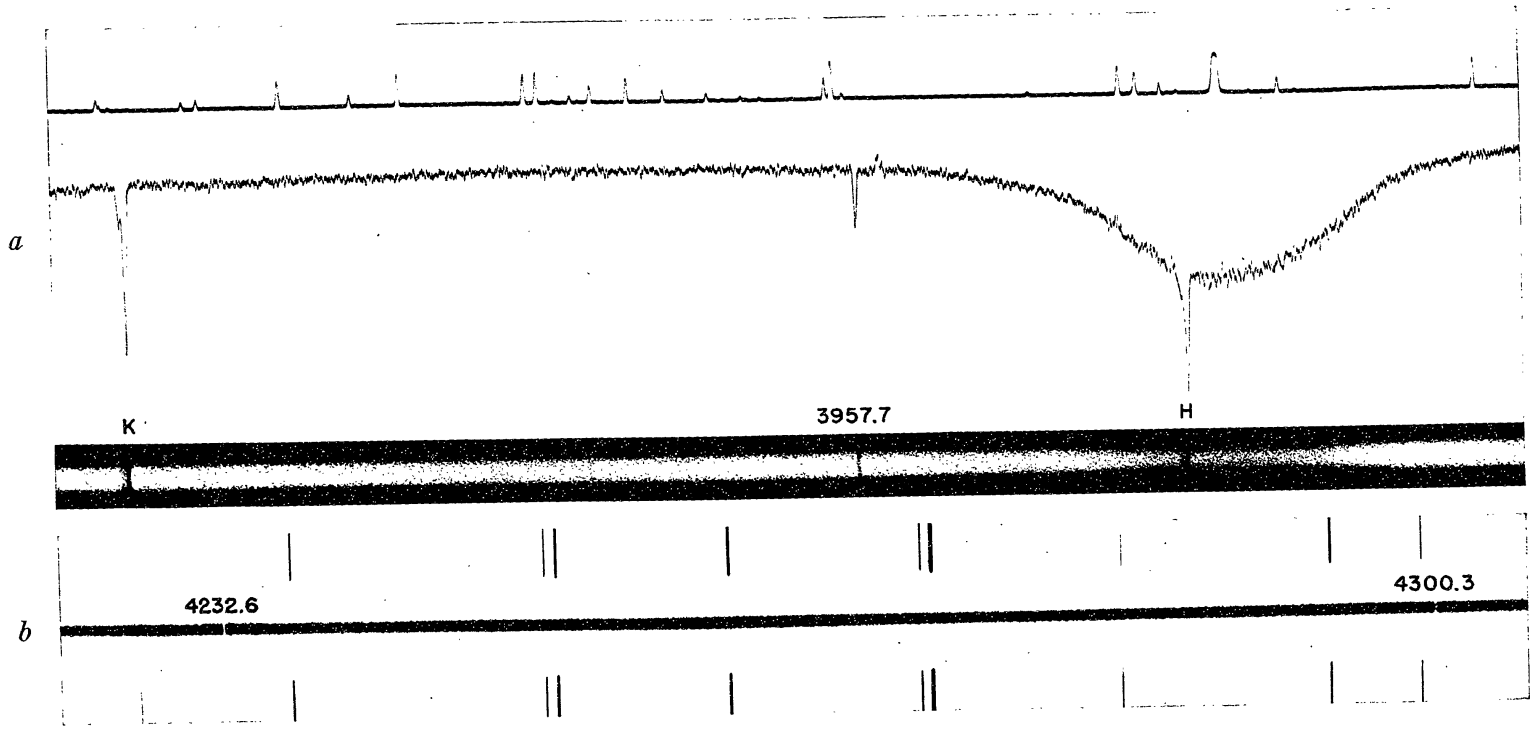
a) α Cygni showing interstellar H and K superposed upon stellar lines; *b*) ζ Ophiuchi, positive reproduction of stellar and comparison spectra, with photometric tracings. Two lines of *CH* are shown, λ 3886 and λ 3890; also λ 3874.6 and a trace of λ 3874.0, both probably *CN*.

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

1941CMWCI.638....1A

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System

PLATE VIII



INTERSTELLAR LINES IN THE SPECTRUM OF ζ OPHIUCHI

a) Interstellar H and K (double) and λ 3957.7 (unidentified); *b*) λ 4232.6 (unidentified) and λ 4300.3 (*CH*)

1941CMWCI.638...1A

λ 4300. An inspection of the stellar spectrogram shows three lines to be present near the predicted positions with relative intensities corresponding to the predicted values. Table 7 gives the results of measurement and a comparison with McKellar's recent values. The lines all belong to the 0, 0 band of *CH*. The measured wave lengths of the lines in ζ Ophiuchi have been corrected for the displacements of interstellar H and K.

The small systematic difference in wave length may readily be accounted for by a difference in the two systems of wave lengths or in the radial velocities of molecular *CH* and atomic *Ca* II. The evidence for the correctness of Dr. McKellar's identification is surprisingly strong and provides definite proof of the existence of interstellar lines arising from molecular transitions.

Assuming McKellar's identification for the lines λ 3874.62 with *CN* and λ 3934.3 with *NaH*, the only remaining narrow unidentified interstellar lines in the region λλ 3800–4700 are λ 3957.72 and λ 4232.58. Both are prominent in the spectrum of ζ Ophiuchi, λ 4232 having about the same intensity as interstellar *H*. Their identification should be greatly facilitated by the results already found.

Since the last paragraph was written, two recent spectrograms of ζ Ophiuchi extending farther to the violet show the presence of two lines not previously observed and one doubtfully suspected but now fully confirmed. These are λλ 3579.04, 3745.330, and 3874.018. The first of these, λ 3579, which is faint, lies nearly an angstrom to the violet of a line of *CN* predicted by McKellar and evidently cannot be identified with it. The line at λ 3745 is fairly prominent. Especial interest attaches to the line λ 3874.018, since this is almost certainly to be identified with λ 3874.00 of *CN* given by McKellar. It is the R(1) line, while the R(0) line is represented by the more conspicuous interstellar line λ 3874.62. The difference in rotational level, as McKellar states, is only 0.00047 volt. Although McKellar's predicted line of *CN* at λ 3579.98 has not as yet been observed with certainty, the close agreement of the two lines at λ 3874 with the two *CN* lines of very low rotational level makes the identification of *CN* in interstellar space highly probable.

I am indebted to Mr. Milton L. Humason both for some of the measures of the interstellar lines and for a careful study of the fainter details upon the spectrograms.

Carnegie Institution of Washington
Mount Wilson Observatory
June 1940

John G. Wolbach Library, Harvard-Smithsonian Center for Astrophysics • Provided by the NASA Astrophysics Data System